Lesson notes · for you

Questions in SQL

Data engineering · Level 3 · Lesson 2

Time	Materials
15 min	Your phone with Darsi, for the SQL and Python steps; Paper and a pen

Learning goals

- Query with SELECT, WHERE, ORDER BY and LIMIT.
- Summarise with COUNT, AVG and GROUP BY; explain NULL.
- Do the same count in Python with a dictionary.

Activity: Human GROUP BY

1. Cards (3 min): Each pupil gets a card for one made-up loan: a class, a book and a date.
2. WHERE (3 min): Only the 5B cards stand up; then they line up newest first: ORDER BY borrowed DESC.
3. GROUP BY (5 min): Everyone groups by class and counts themselves; write one row per class on the board.
4. Check (4 min): Do the group counts add up to the number of cards? Write the SQL for what you just did.

Prepare the loan cards with made-up data before the lesson.

Discussion

- Why do dates written year-month-day sort correctly?
- When do COUNT(*) and COUNT(mark) give different answers?
- How is a Python dictionary like GROUP BY?

Check yourself

- Write a query for the number of loans of each book, the most first.

Try it this week

Write three questions you'd ask your school library's database. Then write each one in SQL: SELECT, FROM, WHERE, and maybe GROUP BY.

Worksheet

Questions in SQL

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Name _____ Class _____ Date _____

1 **Which line keeps only the rows for class 5B?**

◯ ORDER BY class ◯ GROUP BY class ◯ WHERE class = '5B'

2 **Show the pupil and the book for every loan by class 5B, the most recent first.**

```
SELECT pupil, book FROM loans
WHERE class = '5B';
```

Write your program:

3 **WHERE class = 5B, without quotes, finds the rows for class 5B.**

◯ True ◯ False

4 **The librarian's question: which class borrowed the most? Show each class with its number of loans, the most first.**

```
SELECT class, COUNT(*)
FROM loans;
```

Write your program:

5 **Your query says one class borrowed 900 books this term. Which of these are good ways to judge the answer? Choose all that apply.**

☐ Ask whether 900 is even possible for one class

☐ Check that the class counts add up to the total number of loans

☐ Trust it, because it ran without an error ☐ Run it on a few rows you can count by hand

✎ **Write one question you'd ask a database at school (the library, the canteen, the sports team). Which SQL parts would it need?**

Check your answers

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1 **WHERE class = '5B'**

WHERE keeps only the rows that match its condition; the text goes in single quotes.

2

```
SELECT pupil, book FROM loans
WHERE class = '5B'
ORDER BY borrowed DESC;
```

WHERE keeps 5B's five loans, and ORDER BY borrowed DESC sorts them newest first: Grace's 29 September loan comes first.

3 **False**

Without quotes, SQL can't read 5B as text (it isn't a number or a name), so the query stops with an error.

4

```
SELECT class, COUNT(*)
FROM loans
GROUP BY class
ORDER BY COUNT(*) DESC;
```

GROUP BY class makes one row per class, COUNT(*) counts its loans, and ORDER BY COUNT(*) DESC puts the biggest first: 6A with 8, then 5B with 5 and 5A with 4.

5 **Check that the class counts add up to the total number of loans; Run it on a few rows you can count by hand; Ask whether 900 is even possible for one class**

A query that runs can still be wrong: a missing WHERE, or a join that doubles rows.